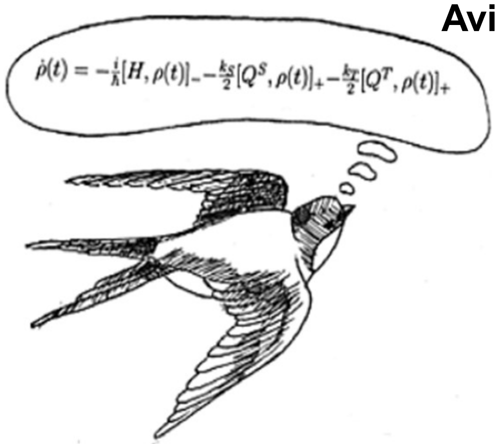
Introduction to Biophysics II: Quantum physics & biology

Lecture 23: Avian compass



Picture from: https://www.ks.uiuc.edu/Research/magsense/ms.html

Reading:
Quantum effects in biology,
M. Moshedi, Y. Omar, GS Engel, M, Plenio
Chapter 10

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How can the magnetic field be perceived?

Problem: The magnetic field of the Earth is very weak, 25-65 μT (1 Tesla = 10^4 Gauss), i.e. ~200 times weaker than a typical permanent magnet, or ~5000 times less than in MRI machine

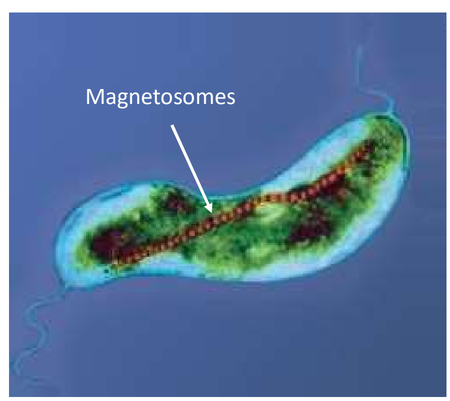
1. Mechanical reception: magnetic particles embedded into a living organism

Example: magnetotactic bacteria.
Organelles (magnetosomes) contain magnetic crystals

Governs *magnetotaxis* – motion along magnetic field lines
They align along magnetic field lines as compass needles (even dead ones)
(precipitate magnetic minerals, like magnetite Fe_3O_4)

Use it to move to intermediate oxygen concentration under water (magneto-aerotaxis)

Note that magnetic lines are not parallel to the surface of water (except on the equator), and moving along magnetic lines, bacteria can change their depth.



https://en.wikipedia.org/wiki/Magnetotactic_bacteria

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How can the magnetic field be perceived?

2. Electric induction: movement in a magnetic field induces an electric field
- Not found in living organisms
- Correction: perhaps works in sharks Nature 2025, (https://www.nature.com/articles/d41586-025-03798-8)

3. Chemical reception:
- Chemical reactions that involve transitions between different spin states can be influenced by magnetic fields so that one of the possible products is favored due to the influence of the magnetic field:
  proposed for birds

Two mechanisms are proposed for birds:

i) Mechanical reception:
Clusters of magnetic iron-containing particles are found in the upper beak, which appear to act as a magnetometer for determining geographical position

ii) Chemical:
quantum spin dynamics of transient photoinduced radical pairs (Schulten 1978, Nature 272, p.85)

Support of (ii) – experimental observation that the compass is light-dependent, birds need UV light
Proposed that free radical chemistry occurs in the bird's retina after excitation of cryptochrome (photoreceptor protein)
Problem: The magnetic Zeeman effect is very weak

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Zeeman effect?

Zeeman effect:
splitting of a spectral line into several components in the presence of a static magnetic field.

In the presence of a magnetic field *B*:

Interaction with magnetic field

$$H = H_0 + V_M$$

Magnetic moment of a molecule

$$V_M = -\vec{\mu} \cdot \vec{B}$$

Total electronic angular momentum

g – Lande g-factor

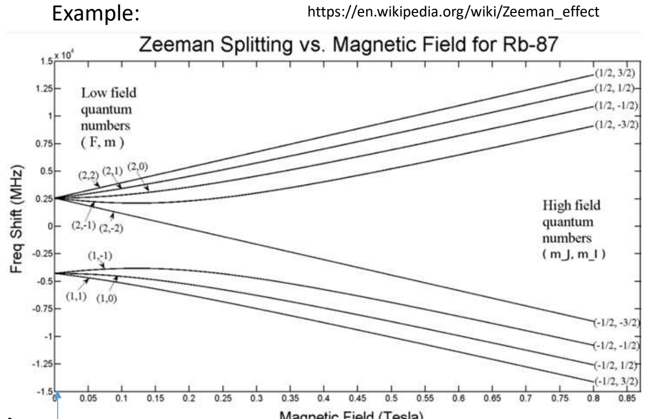
$$\vec{\mu} \approx -\frac{\mu_B g \vec{J}}{\hbar} \qquad \mu_B = \frac{e\hbar}{2m_e} \text{ - Bohr magnetron}$$

Depending on the spin, absorption/emission lines are split (hence names "singlet", and "triplet")

However: $kT_{room} \approx 25$ meV = 6 THz (6e6 MHz)

Zeeman splitting is >6 orders of magnitude smaller than kT

Example: https://en.wikipedia.org/wiki/Zeeman_effect



50μT, Earth's field

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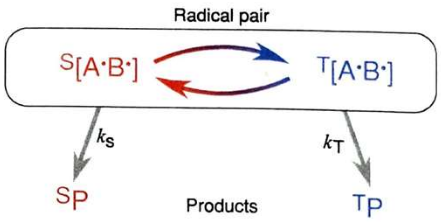
Radical pair: coherent quantum dynamics

The radical pair mechanism is the only well-established way in which an external magnetic field can influence a chemical reaction



Reminder:
Fine splitting: due to the magnetic interaction of electronic spins and orbital angular momentum
Hyperfine splitting: due to electron spin interaction with the nuclear magnetic moment

Magnetic field effect (MFE):

1) A pair of radicals A•B• is formed (charge transfer) in an *entangled* state of singlet (S = 0) or triplet (S = 1), the initial state depends on the spin of the precursor molecule (spin is conserved in the reaction)
2) The two possible states, singlet and triplet, will oscillate (von Neuman-Liouville equation) due to (magnetic) hyperfine interaction with the magnetic field of nuclei ***and external magnetic field***
3) The radical pair can *recombine* from both the S and T states to form chemically distinct products ^{S}P and ^{T}P
4) The fraction of each product depends on k_S, k_T, ***and* the** timing of S←→T conversion of the radical pair

- The yields of the two products and the lifetime of the radical pair become magnetic field-dependent
- Interaction strength depends on the direction between electron spins and the magnetic field

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Radical pair: coherent quantum dynamics

The frequency of quantum beats between two states depends on the interaction strength

- For radical pairs typically: 10^7-10^9 Hz (period 1-100 ns, compare to electron transfer in reaction centers ~1 ps)
- Decoherence – loss of spin correlation – can be significantly slower (>100 ns)

→ There is ample time for the weak magnetic interaction to affect spin dynamics before recombination

Experimental observation:
- Quantum beats in the recombination luminescence of radical ion pairs in non-polar solvents (Grigoryants et a., 1995; Bagryanskyet al, 2000, 2007)
- Detection by EPR (electron paramagnetic resonance) of coherences in radical pairs in photosynthetic reaction centers (Bittl and Kothe, 1991; Kothe et al., 1991; Wang et al., 1992; Dzuba et al., 1996).

Note: They used a strong magnetic field to observe the *magnetic field effect (MFE)*.

Under low magnetic fields, a somewhat different mechanism, the *low field effect (LFE),* comes into play

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LFE: Low Field Effect

Low field effect (LFE) (following Timmel et al., 1998, MOLECULAR PHYSICS, VOL.95, No. 1, 71-89

electron spin interaction with nuclear magnetic moment

Hamiltonian: $\hat{H} = \hat{H}_{HF} + \hat{H}_Z$ (time-dependent sum of hyperfine and Zeeman interactions)

Assume the radical pair is produced in a pure state S (singlet) at time t=0
Density operator of the spin system:

$$\hat{\rho}(0) = \frac{\hat{P}^S}{M} = \frac{1}{M}\sum_{\nu=1}^{M} |S\nu\rangle\langle S\nu|$$

M –number of nuclear spin configurations S_ν
P^S – singlet projection operator

Note: In general, neither H_{HF} nor H_Z commutes with P^S, so the spin system is formed in a non-stationary state
The probability of finding a radical pair in a singlet state at a later time t:

$$\langle \hat{P}^S\rangle(t) = \mathrm{Tr}[\hat{P}^S\hat{\rho}(t)] = \mathrm{Tr}[\hat{P}^S e^{-i\hat{H}t}\hat{\rho}(0)e^{i\hat{H}t}]$$
$$= \frac{1}{M}\mathrm{Tr}[\hat{P}^S e^{-i\hat{H}t}\hat{P}^S e^{i\hat{H}t}]$$

Recall density matrices (von Neuman):
$$\langle A(t)\rangle = \sum_n (\rho A)_{nn} = Tr(\boldsymbol{\rho}\mathbf{A})$$

4 spin states for 2 electrons

Or: $$\langle \hat{P}^S\rangle(t) = \frac{1}{M}\sum_{m=1}^{4M}\sum_{n=1}^{4M}|P^S_{mn}|^2\cos(\omega_{mn}t)$$

Where: $$P^S_{mn} = \langle m|\hat{P}^S|n\rangle$$
$$\omega_{mn} = \langle m|\hat{H}|m\rangle - \langle n|\hat{H}|n\rangle$$

As we had for von Newman – S←→T interconvert with frequency ω_{mn} is defined by the energy difference between states

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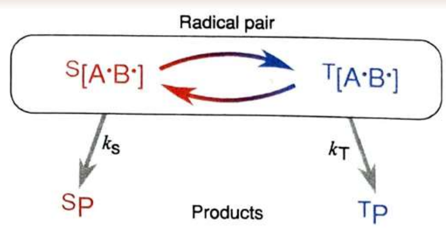
LFE: Low Field Effect

Low field effect (LFE) (following Timmel et al., 1998)
Note: these are crude approximations, just to demonstrate the effect…
Assume that k_S=k_T=k the yield of reaction into ^{S}P state is the given by:



$$\Phi_S = k\int_0^\infty \langle \hat{P}^S\rangle(t)\, e^{-kt}\, dt = \frac{1}{M}\sum_{m=1}^{4M}\sum_{n=1}^{4M}|P^S_{mn}|^2 f(\omega_{mn})$$
$$= \frac{1}{M}\sum_{m=1}^{4M}\sum_{n=1}^{4M}|P^S_{mn}|^2 \boxed{\frac{k^2}{k^2+\omega^2_{mn}}}$$

(Lorentzian function)

Integral used:
$$\int_0^\infty \exp(-k\,t)\cos(w\,t)\,dt = \frac{k}{k^2+w^2} \quad \text{for } k>0$$

The contribution of coherence to singlet yield is determined by its oscillation rate (frequency) relative to the recombination rate.

For k>>|ω_{mn}| → f ≈ 1 , Φ_S ≈ 1
For a significant effect on Φ_S must have k ≤ |ω_{mn}| for at least some (m,n), so that there is time for coherent spin dynamics to convert S to T state.
The frequency ω is defined by splitting in a magnetic field, for an electron in B~50μT will have f ~ 1.4MHz. The effect on Φ_S will become visible for recombination times >100 ns.
Also, random spin dephasing (spin relaxation) must also be >100 ns for coherence to persist

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Requirements for radical-pair compass

(1) The radical pair must be formed in a coherent superposition of its electron-nuclear spin states and at least one of the S and T states should undergo a spin-selective reaction that the other cannot.

(2) There should be suitable anisotropic hyperfine interactions.

(3) The lifetime of the radical pair must be long enough to allow the weak magnetic field to affect the spin dynamics, and the rate constants k_S and k_T should not be too dissimilar.

(4) The Zeeman interaction can only modulate the S→T interconversion if inter-radical spin-spin (exchange and dipolar) interactions are sufficiently weak.
- The molecules in the radical pair should be sufficiently separated

(5) To deliver directional information, the radical pairs must be aligned and immobilized, and the spin system should relax sufficiently slowly

All these factors have been investigated (experiment and/or theory)
→ no obvious obstacles for the mechanism to be possible in cryptochromes
But, also, no direct proof so far…

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Cryptochrome

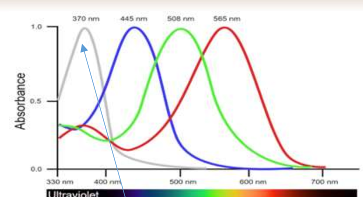
Ritz et al. (2000):
Radical pairs formed photochemically in the protein ***cryptochrome*** could form the basis of the compass magnetoreceptor.

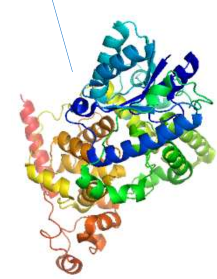
Cryptochromes occur in several of the organisms for which magnetic field effects have been reported (like fruit flies, plants, and migratory birds) and have been shown to act as photoreceptors:

Plants – serve as photosensors (govern growth, timing of flowering, etc.)

Insects – circadian photoreceptor (sleep-wake cycle)

Birds (retina) - ?avian compass?





From greek κρυπτός χρώμα, "hidden color“

By Brautigam, C.A., Smith, B.S., Ma, Z., Palnitkar, M., Tomchick, D.R., Machius, M., Deisenhofer, J. - Protein Data Base, Public https://commons.wikimedia.org/w/index.php?curid=5242162

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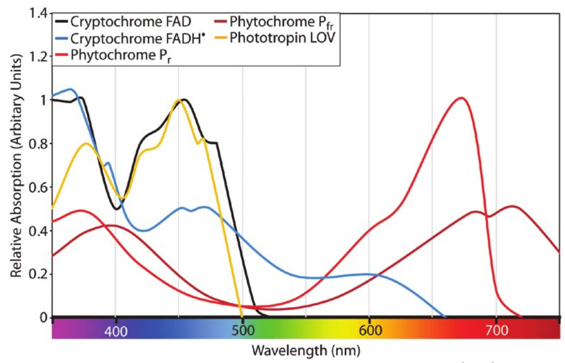
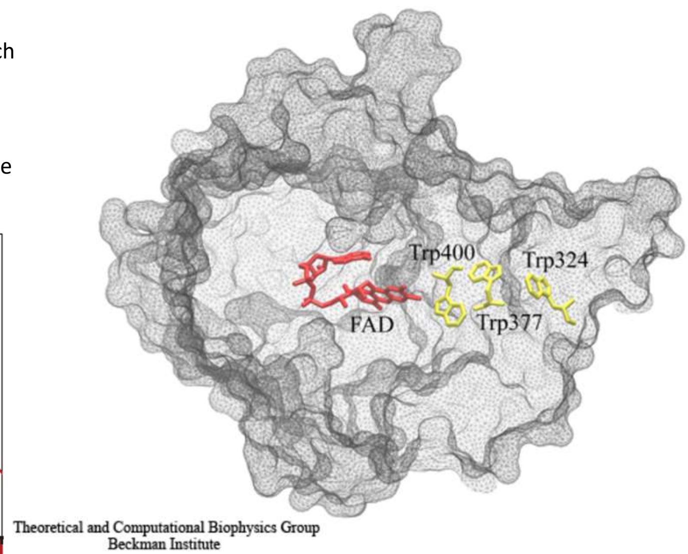
Cryptochrome

Cryptochrome:

Binds FAD (flavin adenine dinucleotide) cofactor, which governs the functioning of the protein.

Light-induced cryptochrome signaling appears to proceed via electron transfer involving a chain of three tryptophan amino acids (the Trp-triad) and FAD.





Battleet al. Journal of experimental botany. 71. 10.1093/jxb/eraa312.

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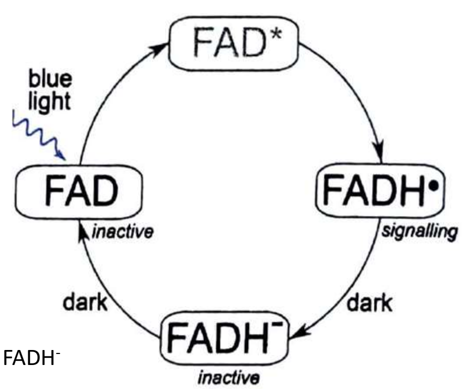
Cryptochrome cycle

FAD can exist in 3 different *redox* forms:
FAD – the neutral, *resting state in the dark*
$FAD^{\bullet}$ (or $FAD^{\bullet-}$) - oxidized
$FADH^{-}$ - protonated oxidized (or $FADH_2$)

Signaling cycle:

1) Excitation of FAD (blue light)
2) Formation of FAD excited state
3) Electron transfer Trp-triad – formation of $FADH^{\bullet}$ radical
   - This state is thought to be responsible for signaling
   - its yield defines the signaling level
4) After milliseconds in the dark, it recombines with an electron, forming $FADH^{-}$
5) And the returns to FAD



Reactions that can be affected by a magnetic field:

$$FAD \rightarrow FADH^{\bullet} \text{ and } FADH^{\bullet} \rightarrow FADH^{-}$$

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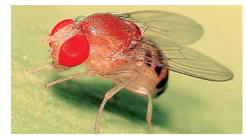
Experimental evidence

Experimental evidence of magnetosensitivity:

- Growth of *Arabidopsis thaliana* seedlings in a 500 uT magnetic field has been reported to enhance cryptochrome activity, such that the plants responded as though they had been exposed to higher intensities of blue light than was, in fact, the case (Ahmad et al., 2007). *Caution:* was not replicated…
- circadian clocks of fruit flies in which cryptochrome acts as a photoreceptor (Yoshii et al., 2009) – an increase in the circadian cycle under blue light *and* a magnetic field. No such effect in cryptochrome knock-out mutants
- *Drosophila melanogaster* (fruit) flies were trained to associate the magnetic field with a food source, and learned to use it as an orientational cue; these responses were absent in cryptochrome-deficient flies. (Gegear et al., 2008, 2010)
- some genetic indications of involvement of cryptochromes in magnetoreception in birds (Freire et al., 2008), but the lack of transgenic birds has precluded more clear-cut evidence



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Experimental evidence

Structurally plausible?

1) Edge-to-edge distance between FAD and *terminal* Trp 324 of the triad is 14.7 Å

Moser-Dutton (Lecture 16-18) (or Marcus):

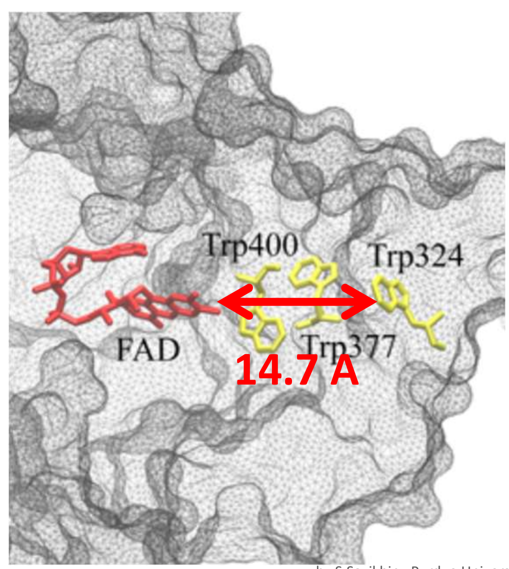
$$\log_{10} k_{ET} \approx 13 - 0.6\left[\mathring{A}^{-1}\right]\left(R_{DA} - 3.6\left[\mathring{A}\right]\right) - 3.1\left[eV^{-1}\right]\frac{\left[\Delta G_0 + \lambda\right]^2}{\lambda}$$

→ The expected electron transfer time is ~1 μs, sufficient for the expected effect.

2) Magnetic response pair should be formed within <1 ns to be in a *pure* spin state: FAD and TRP factors are separated by ~10 Å, sufficient to fast (sequential ET)

3) The magnetic transduction mechanism *can be linked directly to rhodopsin-mediated visual sensing*

→ A bird could *literally see* a visual representation of the Earth's magnetic field



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Anisotropic magnetic field effect

Consider MFE/LFE dependent on magnetic field direction with respect to the molecular coordinate frame:
The relevant coherence P^{S}_{mn} *will* depend on the direction of the magnetic field

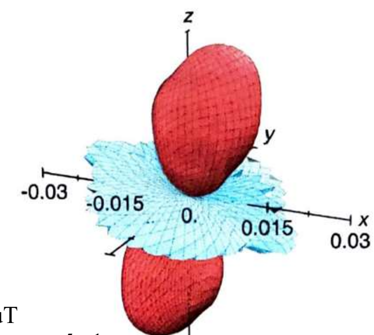
Consider the simple case ((Timmel et al., 2001; Cintolesi et al., 2003):
- one radical has no magnetic nuclei and-
- the other has an axial anisotropic hyperfine interaction with a single spin-I/2 nucleus.
- Assume $k \ll \|\hat{H}_{Z}\| \ll \|\hat{H}_{HF}\|$

Angle between magnetic field and symmetry axis of the hyperfine tensor

Can derive: $$\Phi_S = \frac{1}{4} + \frac{1}{8}\cos^2\psi = \frac{7}{24} + \frac{1}{24}(3\cos^2\psi - 1)$$

Φ_S = ¼ …3/8.   In zero field: Φ_S = 3/8

A more realistic case: simulation of the anisotropic part of Φ_S of FADH$^{\bullet-}$Trp$^{\bullet}$ multinuclear pair (Cintolesi et al., 2003):
Red and blue represent $\Delta\Phi_S$ values that are respectively larger and smaller than the spherical average.



B_0 = 50 uT
$k_S = k_T = 2 \times 10^5\ s^{-1}$.

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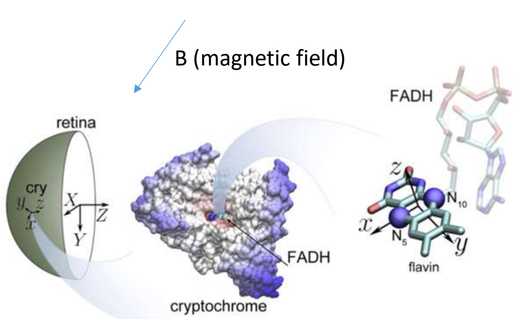
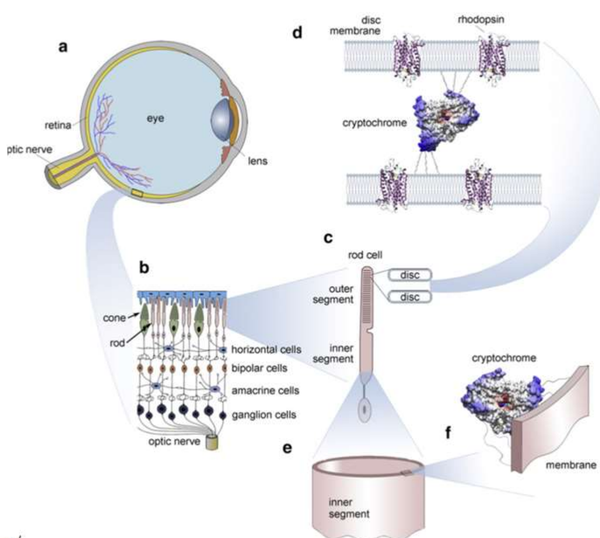
“Visual” magnetic field?

Can birds SEE magnetic field?

Cryptochromes in the retina may be oriented on a spherical surface and will experience different relative directions of the magnetic field, and modulate light sensitivity



$$\langle F(\alpha,\beta,\Theta,\Phi)\rangle = \frac{\int_0^{\pi}\sin\beta'\,d\beta'\int_0^{2\pi}\exp\left[-\left(1-C(\theta')\right)/\sigma^2-\varepsilon_0\sin^2\beta'\right]d\alpha'}{2\pi\int_0^{\pi}\exp\left[-\varepsilon_0\sin^2\beta'\right]\sin\beta'\,d\beta'},$$



Solov'yov, Henrik Mouritsen, Klaus Schulten, Acuity of a Cryptochrome and Vision-Based Magnetoreception System in Birds, Biophysical Journal, Volume 99, Issue 1, 2010, Pages 40-49,

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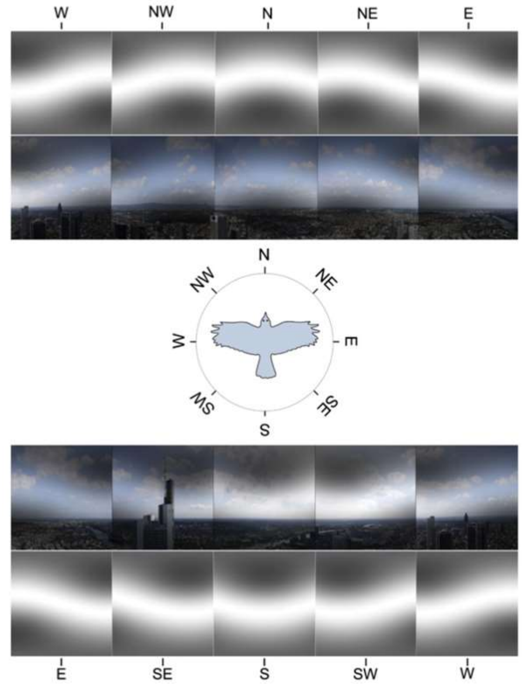
“Visual” magnetic field?

Can birds SEE magnetic field?

FIGURE 5 Panoramic view at Frankfurt am Main, Germany. The image shows the landscape perspective recorded from a bird flight altitude of 200 m above the ground with the cardinal directions indicated (images of Frankfurt provided by Vita Solovyeva). The visual field of a bird is modified through the magnetic filter function according to Eq. 12 with $\eta = 1$ and $\varepsilon_0 = 100$. For the sake of illustration we show the magnetic field-mediated pattern in grayscale alone (which would reflect the perceived pattern if the magnetic visual pathway is completely separated from the normal visual pathway) and added onto the normal visual image the bird would see, if magnetic and normal vision uses the same neuronal pathway in the retina. The patterns are shown for a bird looking at eight cardinal directions (N, NE, E, SE, S, SW, W, and NW). The geomagnetic field inclination angle is 66°, being a characteristic value for the region.



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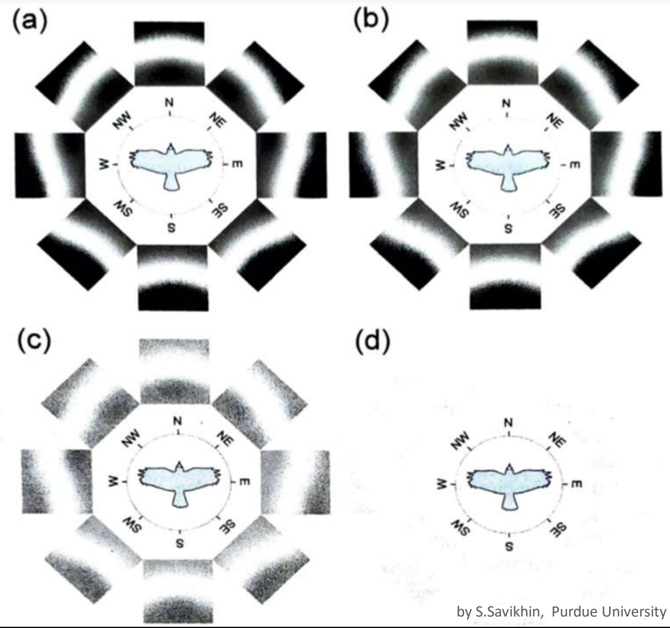
“Visual” magnetic field?

Can birds SEE magnetic field?

Schematic illustration of the visual modulation patterns that might be induced by the geomagnetic field for a bird flying in the eight cardinal directions
(N, NE, E, SE, S, SW, W and NW).

The modulation patterns are calculated for different degrees of orientational disorder of the cryptochrome molecules within the receptor cells, characterized by the parameter ε_0: and magnetic field inclination 66^0

(a) $\varepsilon_0 = 100$ (highly ordered)
(b) $\varepsilon_0 = 10$
(c) $\varepsilon_0 = 3$
(d) $\varepsilon_0 = 1$ (moderately disordered)
(Solov'yov et al., 2010)



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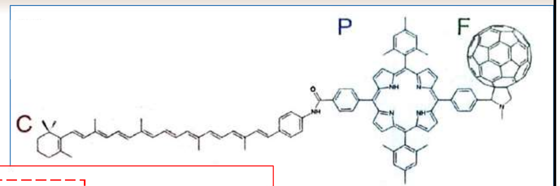
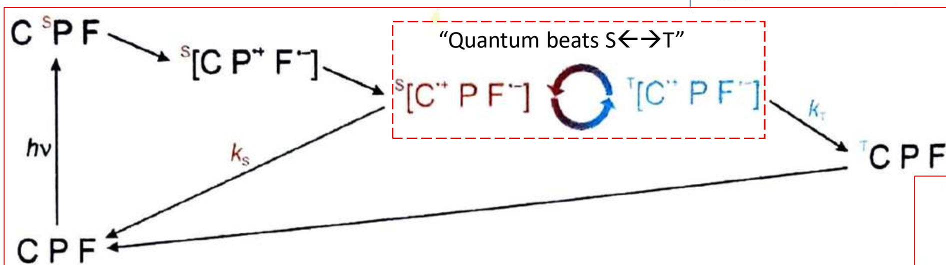
Anisotropic magnetic field effect

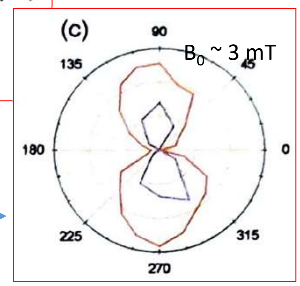
Proof-of-principle experiment (Maeda et al., 2008):
a carotenoid-porphyrin-fullerene triad molecule (CPF)



Reaction:



Porphyrin (similar to Chl) is excited into the singlet state (^SP)
Charge is transferred into two entangled states
The product could be either the triplet state ^{T}CPF, or the original CPF (^{T}CPF has a long lifetime)



Anisotropy of the magnetic field effect on the transient absorption of [$C^{\bullet+}$ P $F^{\bullet-}$] on oriented sample (red), T = 193K

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Radiofrequency (rf) magnetic field effects

Radical pair reaction can be affected by rf fields:
An oscillating *magnetic* field B(t) of an EM wave can modify spin dynamics if close to resonance with the two levels separated due to Zeeman and/or hyperfine coupling.

Example:
Photoinduced inter-molecular electron transfer between chrysene-d_{i2} and 1,4-dicyanobenzene

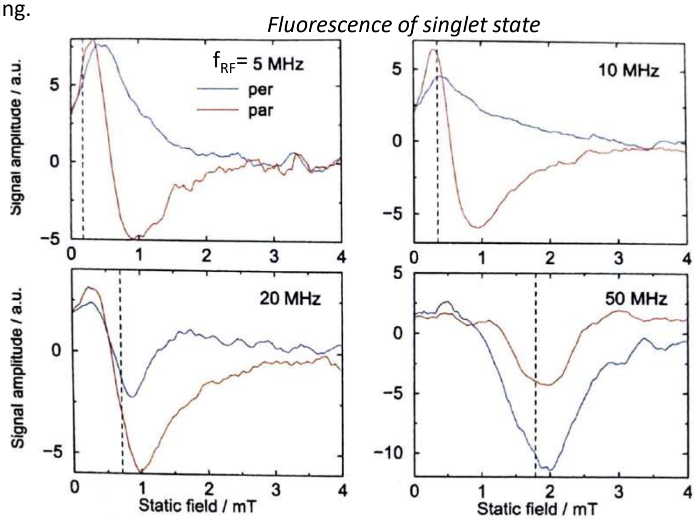
Magnetic and rf radiofrequency (300 μT) are applied simultaneously with polarization of rf field parallel and perpendicular to static field.

Singlet state yield is monitored via fluorescence

The dashed vertical lines: static fields at which the RF radiation is in resonance with the electron Zeeman interaction

→ **Radical pair reactions that exhibit responses to a static magnetic field should also be sensitive to the frequency and direction of an additional RF field**

Fluorescence of singlet state



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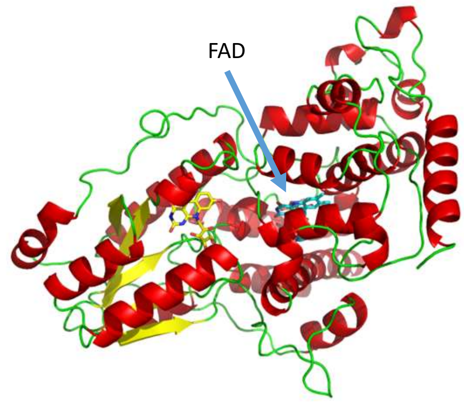
MFE in proteins?

Henbest et al, 2008:
A study of E. coli DNA photolyase, a protein with high sequence homology to cryptochrome and similar photochemistry.

Under light, Flavin and tryptophan radicals similar to those produced in cryptochrome

In the presence of a magnetic field of 39 mT, radical yields changed 3-4% at 5 °C



By Opabinia regalis - Self-created from PDB ID 1QNF using PyMol, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=1783509

Sensitivity of cryptochrome to magnetic field ~50 μT has not yet been demonstrated…
It has been speculated that in the cell response could be significantly larger than in vitro…

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Evidence of radical pair mechanism in birds

Two competing models:
1) iron oxide crystals (conventional)
2) Radical pair, LFE

1) Iron oxide structures are found in beaks near the ophthalmic nerve
   However:
   - magnetic orientation responses are unaffected by anesthetization of the beak or by disabling the ophthalmic nerve.
   - Magnetic orientation response was unaffected by re-magnetizing the magnetic material in a beak with a strong magnetic pulse

   → Birds can detect the direction of a magnetic field without using the iron oxide system in the beak.
   → Another mechanism should exist

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Evidence of radical pair mechanism in birds

Effect of radiofrequency on bird orientation

Experiment (Ritz et al., 2004, 2009):

- Birds were placed into funnel-like cages illuminated by diffuse light and rf field was applied
- Scratch-sensitive paper recorded when birds moved
- Radial density of scratches was an indication of orientation sensitivity

Results:

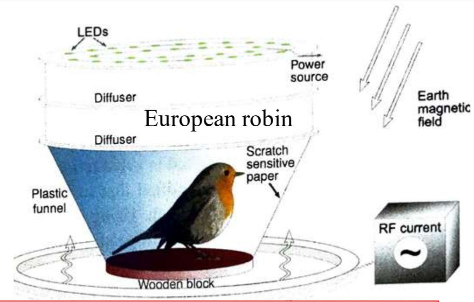
Statistically significant orientation showed using (+)

→ Oscillating fields disrupt the orientation of European robins at frequencies between 0.65 to 7 MHz

→ Analysis suggests that radical pair lifetime or the spin relaxation time, whichever is shorter, is in the range 2-10 µs

→ strong disruptive effect at 1.3135 MHz, corresponding to the spin-only (i.e. g = 2) electron Larmor frequency in the geomagnetic field with only 15 nT intensity, while 480nT was needed at other frequencies

Similar experiments were reproduced in several other species



Intensity (nT)	46 µT static field						92 µT static field	
480	+	+	−	−	−	−		
150			+		+		+	
48				−	+		+	−
15				−	+			−
5				+				+
	0.01	0.03	0.65	1.315	2.63	7.0	1.315	2.63

Frequency (MHz)

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